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Table of Contents

ACKNOWLEDGEMENTS

CHAPTER FIVE - On Unification

CHAPTER SIX - Geometries other than Euclid’s

CHAPTER SEVEN - Algebra without Arithmetic

CHAPTER EIGHT - Matrix Algebra

CHAPTER NINE - Determinants

CHAPTER TEN - Projective Geometry

CHAPTER ELEVEN - On Apparent Impossibilities

CHAPTER TWELVE - On Transformations

CHAPTER THIRTEEN - Finite Arithmetics and Geometries

CHAPTER FOURTEEN - On Groups

ACKNOWLEDGEMENTS

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CHAPTER ONE

On Beauty and Power

No mathematician can be a complete mathematician unless he is also something of a poet.

K. Weierstrass

Wisdom is rooted in watching with affection the way people grow.

Confucius

This is a book about how to grow mathematicians. Probably you have no intention of trying to grow mathematicians. Even so, I hope you may find something of interest here. I myself have no intention of growing plants. I never do any gardening if I can possibly get out of it. But I like very much to look at gardens other people have grown. And I am still more interested if I can meet a man who will explain to me (what very few gardeners seem able to do) just how a plant grows; how, when it is a seed under the earth, it knows which way is up for its stem to grow, and which way is down for the roots; how a flower manages to face towards the light; what chemical elements the plant needs from the soil, and just how it manages to rearrange them into its own living tissue. The interest of these things is quite independent of whether one actually intends to go out and do some hoeing.

What I am trying to do here is to write not from the viewpoint of the practical grower, but for the man who wants to understand what growth is. I am not writing for the professional teacher of mathematics (though teachers may be able to make practical applications of the ideas given here) but for the person who is interested in getting inside the mind of a mathematician.

It is very difficult to communicate the things that are really worth communicating. Suppose, for instance, that you have spent some years in a certain place, and that these years are particularly significant for you. They may have been years of early childhood, or school days, or a period of adult life when new experiences, pleasant or unpleasant, made life unusually interesting. If you revisit this place, you see it in a special way. Your companions, seeing it for the first time, see the physical scene, a pleasant village, a drab town street, whatever it may be. They do not see the essential thing that makes you want to visit the place; to make them see it, you need to be something of a poet; you have to speak of things, but so as to convey what you feel about those things.

But such communication is not impossible. Generally speaking, we overestimate the differences between people. I am sure that if one could go and actually be somebody else for a day, the change would be much less than one anticipated. The feelings would be the same, but hitched on to different objects. Most human misunderstandings are due to the fact that people talk about objects, and forget the varying significance the same object can have for different people.

Generally speaking, teaching conveys thoughts about objects rather than living processes of thought. Suppose someone comes to me with some kind of puzzle; it may be a question in a child's arithmetic book, or a serious problem of scientific research. Perhaps I succeed in solving the puzzle. Then it is quite easy to explain the solution. Suppose I do so; I have shown the questioner how to deal with that particular problem. But if another problem, of a different kind. arises, I shall be consulted again. I have not made my pupil independent of me. What would be really satisfactory would be if I could convey, not simply the knowledge of how to solve a particular puzzle, but the living attitude of mind that would enable my pupil to attack puzzles successfully without help from anyone.

Naturally, there must be certain limitations to what one can expect. Intelligence is one of the factors in problem solving, and it may well be inborn. But there are many other factors - emotions of fear or confidence, habits of self-reliance, initiative, persistence - which depend on education. I do not believe our ancestors at the time of the cave men differed at all from us in inborn qualities of intellect. All historical changes from that time to this, all differences in institutions between one country and another, have essentially been changes in education.1

Pelican Books are themselves a symptom of a profound historical change.

That there should be in so many different countries of the world a large body of thoughtful men and women reading, studying, forming a kind of invisible, international university - a century ago, or in any previous age one would have looked for such a thing in vain.

At the present time, when our knowledge of material things is so great, and our understanding of ourselves so small, a true appreciation of the enormous unused potentialities of education is essential. The industrial revolution implies and requires a psychological revolution. Psychologically, we still belong to the era when people refused to believe that locomotives would run.

The present book was worked out in a country where a great educational change was taking place. In 1948, the University College of the Gold Coast was founded. The students were keen and of first-rate ability. So far as inborn intelligence went, they were capable of becoming, within ten or twenty years, research workers in mathematics, university lecturers, professors. But of course there was no mathematical tradition in the country. That had to be created.

It was therefore necessary to obtain a sort of essence of mathematics ; to examine the life of a budding mathematician in one of the older mathematical centres; to study all the influences that helped him to grow; the atmosphere of school and college, countless hints, allusions, suggestions from older mathematicians and from books. From all of this to try to form some clear idea of what we were trying to do, what the qualities of a mathematician were, how they were to be stimulated.

In the first five chapters of this book, I try to give a specification of what a mathematician is, and how he grows. These chapters also contain various pieces of mathematics to illustrate what interests a mathematician. The remainder of the book is an exposition of various branches of mathematics; these have been selected for their strangeness, their novelty, their stimulating power. Moreover, they are elementary. A confused recollection of School Certificate mathematics should be sufficient to see a reader through them.

[image file=image_rsrc3T2.jpg] Figure 1 - THE STRUCTURE OF THIS BOOK

Chapters 1-5 use mathematics to illustrate the qualities of a mathematician. They are not shown in the diagram.

Where one chapter is shown as resting on another, the upper chapter makes use of the ideas explained in the lower one. Thus, it is necessary to read Chapter 10 before Chapter 11.

An arrow going from one chapter to another indicates some connexion between the two chapters. For instance, one section of Chapter 13 (the section 'Finite Geometries') cannot be understood without reference to Chapter 11. But all the rest of Chapter 13 could be read as the first chapter of the book.

It will be seen that bungalows rather than skyscrapers predominate.

Calculus is referred to once or twice, but is nowhere used as part of the argument. This fact is interesting, as showing that some parts of recent mathematics (i.e. since 1800) are not a development of the older work, but have gone off in quite a new direction.

Nor will you find much in the way of long calculations here. Nearly every mathematical discovery depends upon a fairly simple idea. Textbooks often conceal this fact. They contain massive calculations, and convey the impression that mathematicians are men who sit at desks and use up vast quantities of stationery. This impression is quite wrong. Many mathematicians can work quite happily in a bath, in bed, while waiting at a railway station, or while cycling (preferably not in traffic). The calculations are made before or after. The discovery itself grows from a central idea. It is these central ideas I hope to convey. Naturally, some details must be given, if the book is to be mathematics at all, and not merely a sentimental rhapsody on the mathematical life.

But before you part with your money, I want to warn you that there will be one or two pages on which you may see some lengthy algebraic expression. One of the things I want to discuss is how to look at an algebraic expression. The longer the expression, the more important it is to know how to look at it. For instance, in Chapter 9, on determinants, you will see some long, sprawling chains of symbols. These are only introduced so that we can exclaim, ‘Look at this mess! How are we to discover, in this apparent chaos, some simple idea that we can think about? Where lies the shape, the form, the pattern, without which nothing can be regarded as mathematics?’

A point that should be borne in mind is that, generally speaking, higher mathematics is simpler than elementary mathematics. To explore a thicket on foot is a troublesome business; from an aeroplane the task is easier.

One thing most emphatically this book does not claim to do; that is, to give a balanced account of the development of mathematics since 1800, nor even to give a rounded account of what mathematicians are doing to-day. This book presents samples of mathematics. It does not claim to do more. Indeed, it is doubtful if more could be done in a book of this size.

THE EXTENT OF MATHEMATICS

Very few people realize the enormous bulk of contemporary mathematics. Probably it would be easier to learn all the languages of the world than to master all mathematics at present known. The languages could, I imagine, be learnt in a lifetime; mathematics certainly could not. Nor is the subject static. Every year new discoveries are published. In 1951 merely to print brief summaries of a year’s mathematical publications required nearly 900 large pages of print. In January alone the summaries had to deal with 451 new books and articles. The publications here mentioned dealt with new topics; they were not restatements of existing knowledge, or very few of them were. To keep pace with the growth of mathematics, one would have to read about fifteen papers a day, most of them packed with technical details and of considerable length. No one dreams of attempting this task.

The new discoveries that mathematicians are making are very varied in type, so varied indeed that it has been proposed (in despair) to define mathematics as ‘what mathematicians do’. Only such a broad definition, it was felt, would cover all the things that might become embodied in mathematics; for mathematicians to-day attack many problems not regarded as mathematical in the past, and what they will do in the future there is no saying.

A little more precise would be the definition ‘Mathematics is the classification of all possible problems, and the means appropriate to their solution'. This definition is somewhat too wide. It would include such things as the newspaper's 'Send your love problems to Aunt Minnie', which we do not really wish to include.

For the purposes of this book we may say, 'Mathematics is the classification and study of all possible patterns'. Pattern is here used in a way that not everybody may agree with. It is to be understood in a very wide sense, to cover almost any kind of regularity that can be recognized by the mind. Life, and certainly intellectual life, is only possible because there are certain regularities in the world.2 A bird recognizes the black and yellow bands of a wasp; man recognizes that the growth of a plant follows the sowing of seed. In each case, a mind is aware of pattern.

Pattern is the only relatively stable thing in a changing world. To-day is never exactly like yesterday. We never see a face twice from exactly the same angle. Recognition is possible not because experience ever repeats itself, but because in all the flux of life certain patterns remain identifiable. Such an enduring pattern is implied when we speak of 'my bicycle' or 'the river Thames', notwithstanding the facts that the bicycle is rapidly rusting away and the river perpetually emptying itself into the sea.

Any theory of mathematics must account both for the power of mathematics, its numerous applications to natural science, and the beauty of mathematics, the fascination it has for the mind. Our definition seems to do both. All science depends on regularities in nature; the classification of types of regularity, of patterns, should then be of practical value. And the mind should find pleasure in such a study. In nature, necessity and desire are always linked. If response to pattern is characteristic alike of animal and human life, we should expect to find pleasure associated with the response to pattern as it is with hunger or sex.

It is interesting to note that pure mathematicians, moved only by their sense of mathematical form, have often arrived at ideas later of the utmost importance to scientists. The Greeks studied the ellipse more than a millennium before Kepler used their ideas to predict planetary motions. The mathematical theory needed by relativity was in existence thirty to fifty years before Einstein found a physical application for it. Many other examples could be given.

On the other hand, many very beautiful theories, to which any pure mathematician would concede a place in mathematics for their intrinsic interest, arose first of all in connexion with physics.

NATURE'S FAVOURITE PATTERN?

Another striking fact is that in nature we sometimes find the same pattern again and again in different contexts, as if the supply of suitable patterns were extremely limited. The pattern which mathematicians denote by Δ2V occurs in at least a dozen different branches of science. It arises in connexion with gravitation, light, sound, heat, magnetism, electrostatics, electric currents, electromagnetic radiation, waves at sea, the flight of aeroplanes, vibrations of elastic bodies and the mechanics of the atom - not to mention a pure mathematical theory of first-class importance, the theory of functions f(x + iy), where i is [image file=image_rsrc3T3.jpg]

Practical men often make the mistake of treating all these applications as quite separate and distinct. This is a great waste of effort. We have not twelve theories, but one theory with twelve applications. The same pattern appears throughout. Physically the applications are distinct, mathematically they are identical.

The idea of the same pattern arising in different circumstances is a simple one. One only has to invent a Greek name for this idea to have one of the commonest terms of modern mathematics - isomorphic (isos, like; morphe, shape - having the same shape). Nothing delights a mathematician more than to discover that two things, previously regarded as entirely distinct, are mathematically identical. 'Mathematics', said Poincaré, 'is the art of giving the same name to different things.'

One might ask, ' Why does this one pattern, Δ2V, occur again and again?' Here we tremble on the brink of mathematical mysticism. There can be no final answer. For suppose we show that this pattern has certain properties which make it particularly suitable; we then have to ask, 'But why does Nature prefer those properties?' - in endless mazes lost. Nevertheless a certain answer can be given as to why Δ2V so frequently occurs.3

The impossibility of any final answer to the question, 'Why is the universe like it is?' does not therefore mean that the enquiry is entirely useless. We may succeed in discovering that all scientific laws so far discovered have certain properties in common. A mathematician, in studying what patterns have these properties, has a reasonable belief that his work will be useful to future generations - not a certainty, of course; nothing is certain. He may also hope to satisfy his own desire to achieve a deep insight into the workings of the universe.

THE MATHEMATICIAN AS CONSULTANT

Technical men and engineers do not, as a rule, have the vision of mathematics as a way of classifying all problems. They tend to learn those parts of mathematics which have been useful for their profession in the past. In the face of a new problem they are accordingly helpless. It is then that the mathematician gets called in. (This division of labour between engineers and mathematicians is probably justified; life is too short for the simultaneous study of technical practice and abstract pattern.) The encounter between the mathematician and the technician is usually amusing. The practical man, by daily contact with his machinery, is so soaked in familiarity with it that he cannot realize what it feels like to see the machine for the first time. He pours out a flood of details which, to the consulting mathematician, mean precisely nothing. After some time, the mathematician convinces the technical man that mathematicians are really ignorant, and that the simplest things have to be explained, as to a child or to Socrates. Once the mathematician understands what the machine does, or is required to do, he can usually translate the problem into mathematical terms. Then he can tell the practical men one of three things, (i) that the problem is a well-known one, already solved, (ii) that it is a new problem which perhaps he can do something about, (iii) that it is an old problem which mathematicians have tried to solve without success, and that several centuries may elapse before any advance is made with it; the factory must deal with the problem empirically. Situation (iii) occurs with distressing frequency. But situations (i) and (ii) do also occur, and it is then that the mathematician, by his interest in and classification of patterns, can be of service to trades and professions about which, in one sense, he knows nothing.

A mathematician who is interested in consultative work therefore needs not merely to study problems which have occurred, he must be prepared for those that may occur. To a certain extent, one can recognize that practical problems form a type. For example, very often a practical problem takes the form of a differential equation.4 Some differential equations we know how to solve, others we do not. A mathematician may therefore seek to enlarge his armoury by studying those differential equations which have so far defied solution. And this will lead him to ask all sorts of fundamental questions, such as, 'What is the difference between the equations that have been solved and those that have not? What makes an equation easy or difficult to solve?'

But practical problems do not always conform to type. Sometimes problems arise which are quite unlike those of normal routine. The clue to their solution may be found quite by chance; perhaps they resemble some puzzle solved in an idle moment. When this happens, the puzzle may prove to be the foundation of a new and dignified mathematical theory. This doctrine, like all doctrines, is capable of abuse. A man may waste his life on footling little puzzles, and defend this on the ground that these might be the beginnings of new branches of mathematics. So they might: the matter depends on one's judgement of what is likely to prove important, and there is no rule by which to settle the question. And any mathematician will agree that there are subjects which have not yet found any technical application, but which one feels to be major parts of mathematics. They are part of the battle, they are not escapism. Some time in the future like the ellipse they will find their Kepler, like tensor analysis their Einstein. But in any case there they are all ready, massive machines to solve a certain class of problem, if the necessity arises.

THE MATHEMATICIAN AS ARTIST

As I write this, I imagine a pure mathematician reading it with increasing distress; that is, supposing he gets as far as this. 'You are treating mathematics,' he will be saying, 'as something useful. But mathematics is not a means to something else; it is an end in itself. Not the usefulness of mathematics is important but the beauty. Technical mathematics is the dullest part of mathematics. Look at the people working on the theory of numbers, which has no application at all; would you prefer them to be working at bookkeeping?'

This view - which is sincerely held by many eminent and some great mathematicians - may be contrasted with its opposite, the utilitarian, bureaucratic view of mathematics. According to this somewhat Puritanical theory, mathematicians should be ashamed of their interest in beauty and elegance; they should only work at mathematics when some official summons them to solve an immediately useful problem.

Both views are incomplete. Either of them, pursued to a logical conclusion, would be fatal to mathematical and even to technical progress.

Let us examine the ‘mathematics for mathematics’ sake’ theory first and consider (if I may take a particular example) its application to the Gold Coast. The Gold Coast is a country of lively, intelligent, and cheerful people; let nothing here said suggést that it is a miserable place. But it is a country with certain acute material needs. In many places water supply is uncertain, sanitation lacking, disease widespread, food inadequate. Some children grow up permanently hungry. How then are we to defend the expense of the mathematics department in the new university? On the grounds of the beauty of mathematics? To defend mathematics in such circumstances purely on the grounds of its beauty is the height of heartlessness. Mathematics has cultural value; but culture does not consist in stimulating oneself with novel patterns in indifference to one’s surroundings. Obviously the power of mathematics, mathematics as a means to engineering and medical science, is of first-rate importance in any developing country; the beauty without the power is futile.

But there is still a word to be said for the artist. Power without beauty is liable to be impotent. An activity engaged in purely for its consequences, without any pleasure in the activity itself, is likely to be poorly executed. An engineer may begin to study mathematics because it is useful for his profession, but if his concern stops there, if he does not begin to feel the fascination of the subject itself, he will not do much good with mathematics.

The utilitarian view of mathematics is realistic in one way; it recognizes the fact that a mathematician is a human being, dependent on the efforts of other human beings for his food and clothing and shelter and fuel, and owing some return for these. This is the aspect of the matter most easily appreciated by non-mathematicians, by administrators and by taxpayers. Considerations of utility may show that a country needs mathematicians; but yet give no clue as to how it is to get them. It might be very desirable for the Sahara to be covered by a forest of oak trees, to give shade to travellers, but that does not cause trees to grow there. Mathematicians, like trees, are living organisms and will only grow in conditions where they can grow.

It might be objected that men are not trees; that if a man realizes something ought to be done, he can go and do it. This is true within certain limits. There can be social conditions favourable to mathematical studies; if a country urgently needs mathematicians, and if everyone knows this, mathematics may well flourish. But this still does not answer the question of how it comes to flourish. An external motive, good or bad, is not' enough. Greed for money, desire for fame, love of humanity are equally incapable of making a man a composer of great music. It has been said that most young men would like to be able to sit down at the piano and improvise sonatas before admiring crowds. But few do it; to desire the end does not provide the means; to make music you must be interested in music, as well as (or instead of) in being admired. And to make mathematics you must be interested in mathematics. The fascination of pattern and the logical classification of pattern must have taken hold of you. It need not be the only emotion in your mind; you may pursue other aims, respond to other duties; but if it is not there, you will contribute nothing to mathematics.

To this extent, the artist is more realistic than the bureaucrat. The artist does at least understand how people become mathematicians. Both the pure artist and the pure bureaucrat are wrong, or at least incomplete. If the teaching of mathematics had to be based on the theories of either one of them, that of the artist would do less harm. An artist may be an anarchist, a bohemian, or a tramp, but at least he is alive, and without life there is no growth. If such a man can teach children to love a subject for itself, there is always the hope that at a later age these children will turn their gifts to useful ends. But if they are left in the hands of the extreme utilitarians, they will have no gifts to turn.

CHAPTER TWO

What are the Qualities of a Mathematician?

I do not fancy this acquiescence in second-hand hearsay knowledge; for, though we may be learned by the help of another's knowledge, we can never be wise but by our own wisdom.

Montaigne, Of Pedantry

Mental venturesomeness is characteristic of all mathematicians. A mathematician does not want to be told something; he wants to find it out for himself. An adult mathematician, of course, if he hears that some great discovery has been made will want to know what it is, he will not want to waste time re-discovering it. But I am thinking of'mathematicians at an early age, where this mental aggressiveness is very marked. For example, if you are teaching geometry to a class of boys nine or ten years old, and you tell them that no one has ever trisected an angle by means of ruler and compasses alone, you will find that one or two boys will stay behind afterwards and attempt to find a solution. The fact that in two thousand years no one has solved this problem does not prevent them feeling that they might get it out during the dinner hour. This is not exactly a humble attitude, but neither does it necessarily indicate conceit. It is simply the readiness to respond to any challenge. In fact, of course, the trisection of the angle by the means specified has been proved to be impossible; it is in the same category as trying to express [image file=image_rsrc3T4.jpg] as a rational fraction p/q.

Again, a good pupil will always be running ahead of the course. If you show him how to solve a quadratic equation by completing the square, he will want to know whether it is possible to solve a cubic equation by completing the cube. The rest of the class do not ask such a question. Having to solve quadratics is bad enough for them; they do not wish to add to their burdens.

The desire to explore thus marks out the mathematician. This is one of the forces making for the growth of mathematics. The mathematician enjoys what he already knows; he is eager for new knowledge. Fractional indices, in school algebra, seem to illustrate this point. I can well imagine anyone, after reading an elementary account of fractional and negative indices, wondering whether such things are really justified at all. There are many logical difficulties to be overcome. The discoverer of fractional indices, I feel, must have enjoyed working with ordinary indices and been so anxious to extend this subject that he was willing to take the logical risks. A new discovery is nearly always a matter of faith in the first instance; later, of course, when one has seen that it does work, one has to find a logical justification that will satisfy the most cautious critics.

Interest in pattern has already been mentioned. Pattern appears already in the first steps of arithmetic; in the fact, say, that four stones can be arranged to form a square, while five cannot. Mathematical, like musical, ability is apparent at a very early age; four years old, or even earlier. A young child once said to me, 'I like the word September. It goes sEptEmbEr'. I had never myself noticed this pattern . .. .. * . in the vowels and consonants of 'September'. It is of course perfectly symmetrical. Such a child should enjoy arithmetic.

A very elementary example of pattern is contained in the multiplication tables. Children usually like the 2 × and 5 × tables, because the final digits are easy to remember - always even in the 2 ×, and still better, always 0 or 5 in the 5 ×. But even the 7 × table has its regularities. If one examines the final digits of 7,14, 21, 28, 35, 42, 49, 56, 63, 70 they are

7 4 1 8 5 2 9 6 3 0

with the differences -3 -3 + 7; - 3 - 3 + 7; - 3 - 3 - 3, in which quite a definite rhythm is apparent.5 The final digits of the 7 x table read backwards are of course those of the 3 x table. Even in the lowest forms at school, the habit of observing mathematical regularities can grow. Much of the early work of Gauss springs from his habit of making calculations and observing the results. Hermite, a great French mathematician, also stressed the importance of observation in leading to mathematical discoveries.6 Of course observation alone is not sufficient to make a great mathematician.

Just as a pupil who notices arithmetical regularities will find this helpful in doing arithmetic, so the observation of regularities in algebra helps one to avoid or detect slips.

For example, the condition for the quadratic equation $ax^2 + 2bx + c = 0$ to have equal roots is $b^2 - ac = 0$. (The reader may be more familiar with this in the form $b^2 - 4ac = 0$, which holds for $ax^2 + bx + c = 0$.) One can find a similar condition in connexion with a cubic equation. A cubic equation has three roots; we enquire in what circumstances two of these three roots are equal. For the equation $ax^3 + 3bx^2 + 3cx + d = 0$ the condition is $(bc - ad)^2 - 4(ac - b^2)(bd-c^2) = 0$. If you like, you can multiply out completely and write the condition as

$$a^2d^2 - 6abcd + 4b^3d + 4ac^3 - 3b^2c^2 = 0.$$

If you work with such expressions you can hardly help noticing certain things.

(i) The total number of letters in each term of the expression is the same. For instance, in $b^2 - ac$, each term contains two letters multiplied together, *i.e.* every term is of the second degree. In the longer condition for the cubic to have equal roots, every term contains four letters multiplied together. b^3d is of course short for bbbd. Every term is of the fourth degree.

(ii) It is not so obvious, but there is another kind of balance in these expressions, a balance between letters which occur early in the alphabet and those which occur late. For instance, in the condition for the cubic abcd occurs. The term a^2d^2, or aadd, also occurs. In the term aadd, the second letter a occurs earlier in the alphabet than b, the second letter of abcd. But justice is done. If we look at the third letters, aadd contains d while abcd contains the earlier letter c. The balance is exact; b is immediately after a, c is immediately before d. The same balance holds throughout. This may be checked in the following way. Suppose we let each term score in the following way. a, being the earliest letter in the alphabet, scores 0; b scores 1, c scores 2, d scores 3. It will be found that each term scores a total of 6. For example, abcd scores $0 + 1 + 2 + 3 = 6$; ac^3 scores $0 + 2 + 2 + 2$.

The technical term for this score is weight: we say that each term has the weight 6. In $b^2 - ac$, for a quadratic, each term has weight 2.

(iii) The sum of the coefficients in each expression is zero. In $b^2 - ac$ the coefficients are 1 and -1, which add up to 0. In $a^2d^2-6abcd + 4b^3d + 4ac^3 - 3b^2c^2$ the coefficients are 1, - 6, 4, 4, - 3 with the sum 0. Another way of expressing the same thing is to say that if we put a, b, c, d all equal to 1, the expression takes the value zero.

These three things we have noticed provide a check on our accuracy. (i) enables us to check if we have made a slip in writing the power to which any letter occurs, for such a slip will make the terms of unequal degree (unless of course we make several slips, which compensate each other in this respect). (ii) will save us from slips in copying the letters. If, for example, we copied d as a at some stage of the working, this would alter the weight of a term. Test (iii) will save us from making slips in adding the terms together. Suppose for example that we overlook a term, when we are collecting the terms together. On putting the value 1 for every letter we shall find we do not get zero, and our attention will be drawn to the mistake.

This type of checking does not give a profound insight into mathematics; it is one of the things a mathematician does almost unconsciously, to look at an answer and make sure it has the kind of symmetry, the kind of balance he expects. It is surprising how often pupils are not taught to examine things in some such way. In examinations, one again and again finds pupils handing in answers which - in the phrase of G. K. Chesterton's Father Brown - have 'the wrong shape', and which obviously are crying out for some simple correction. The principle extends far beyond the special type of work discussed above.

A GEOMETRICAL PATTERN

In school geometry one meets the result

[image file=image_rsrc3T5.jpg] where x, a, b, c, d stand for the lengths of the lines shown in Figure 2.

This result is very easily proved, by finding cos Q and cos S in terms of the sides of the triangles PQR and PSR. The sum of the two expressions must be zero, for Q + S = 180° (cyclic quadrilateral), so cos S = - cos Q. The resulting equation is then solved for x^2.

There is nothing very striking in this proof, but the pattern of the result is remarkable.

[image file=image_rsrc3T6.jpg] Figure 2There are three ways of dividing four objects up into pairs. If four men play bridge, A and B are partners against C and D; or A and C against B and D; or A and D against B and C. No other arrangement is possible.

The algebraic expressions ab + cd, ac + bd, ad + bc are also built by dividing up the four symbols a, b, c, d into pairs, and putting a + sign in the middle. These three, and only these three, can be formed in this way; cd + ab of course is the same as ab + cd.

In the formula above all three expressions occur, two of them in the numerator, the remaining one in the denominator.

I do not here want to go into the question why this occurs, but simply to point out that this formula is, in a quiet way, a very memorable one.

SIGNIFICANCE AND GENERALIZATION

In other arts, if we see a pattern we can admire its beauty; we may feel that it has significant form, but we cannot say what the significance is. And it is much better not to try. A poet protested against the barbarous habit of requiring children to paraphrase poems. The only way you can explain the meaning of a poem, he said, is by writing a better poem, and that is a lot to ask of children.

But in mathematics it is not so. In mathematics, if a pattern occurs, we can go on to ask, Why does it occur? What does it signify? And we can find answers to these questions. In fact, for every pattern that appears, a mathematician feels he ought to know why it appears.

For example, we can explain why conditions for equal roots must have the properties (i), (ii), and (iii) described earlier. I will indicate the general idea, and omit the algebraic details (which are actually quite simple).

[image file=image_rsrc3T7.jpg] Figure 3a and b

Let us consider the graph of a cubic, which for short I will call y=f(x). The first graph shown (Figure 3a) represents a cubic with all its roots real and unequal. But if the cubic were lifted up until the points B and C coincided, we should obtain the second graph, which represents a cubic with a pair of equal roots (Figure 3b). In passing it may be mentioned that calculus gives us a simple test for equal roots. At the point E, both

[image file=image_rsrc3T8.jpg] are zero. Equal roots occur if the equation f‘(x) = 0 has a root in common with the original equation f(x) = 0.

Now imagine that the second graph (Figure 3b) were drawn on a sheet of rubber. Suppose this rubber were to be stretched vertically; this would have the effect of changing the scale on the y-axis. The graph would become that of y = kf(x). Now obviously if the original graph touched the x-axis (which it does when there are equal roots), the new graph, obtained by stretching, would still do so. That is to say, if f(x) = 0 has equal roots, so has kf(x) = 0. By considering the effect of the extra factor k on the constants a, b, c, d, the coefficients in f(x), one can show that the condition for equal roots must have all its terms of equal degree - property (i).

In the same way, stretching horizontally will not affect the general appearance of the graph. It will still touch the x-axis. From this we may conclude that if f(x) = 0 has equal roots, so has f(kx) = 0. This leads us to property (ii), that all the terms have equal weight.

Property (iii) is the simplest of all. If we replace all the letters a, b, c, d by 1, the quadratic equation becomes x2 + 2x + 1 = 0 and the cubic x3 + 3x2 + 3x + 1 = 0. These equations are (x + 1)2 = 0 and (x + 1)3 = 0, which obviously have the root -1 1 repeated. (All three roots of the cubic are -1; it would have been sufficient for our purpose if two of them had been -1.)

So properties (i) and (ii) hold for any condition that is unaffected by a change of scale of y (a vertical stretch) and a change of scale of x (a horizontal stretch); property (iii) holds for any condition that is satisfied by (x + 1)n = 0.

This represents a great advance on the time when we had simply noticed the properties (i), (ii), (iii). We have now interpreted them; we know why they hold and when they hold.

And this allows us to apply tests (i), (ii), (iii) to types of condition other than those for equal roots. For example, with a cubic equation, which has three roots, we may ask in what circumstances one root occurs exactly mid-way between the other two. In the first cubic graph, drawn earlier, this would mean the point B is to be half-way from A to C. This property would not be destroyed by changes of scale, vertical or horizontal. This property holds for the equation $(x + 1)3 = 0$; for this equation, the points A, B, C all coincide at $x = -1$, and B thus is the mid-point of AC. The algebraic condition for this property must therefore have the features (i), (ii), (iii). The condition is in fact $2b3 - 3abc + a2d = 0$, and as you can see it is of the form expected.

This is an example of generalization, one of the most important factors in the development of mathematics. We began with an observation which applied only to conditions for equal roots; we ended with a principle which applied to a much wider class of conditions. This is obviously valuable; the wider the situations to which a principle is relevant, the more often it is likely to help us out of our difficulties. As Poincaré said, ‘Suppose I apply myself to a complicated calculation and with much difficulty arrive at the result, I shall have gained nothing by my trouble if it has not enabled me to foresee the results of other analogous calculations, and to direct them with certainty, avoiding the blind groping with which I had to be contented the first time’.[7]

GENERALIZATION AND SIMPLICITY

When we generalize a result, we make it more useful. It may strike you as strange that generalization nearly always makes the result simpler too. The more powerful result is easier to learn than the less powerful one.

This may be illustrated by means of a very trivial puzzle, which runs as follows: One glass contains ten spoonfuls of water. Another glass contains ten spoonfuls of wine. A spoonful of water is taken from the first glass, put into the second glass, and the mixture is thoroughly stirred up. A spoonful of this mixture is then transferred to the first glass. Will the amount of wine in the first glass, at the end of this procedure, be more or less than the amount of water in the second glass?

The obvious way to go about this question involves the following calculation. After a spoonful of water has been put into the wine, the second glass contains 10 spoonfuls of wine and 1 of water, 11 spoonfuls in all. One spoonful of this mixture will therefore contain [image file=image_rsrc3T9.jpg] spoonful of wine, [image file=image_rsrc3TA.jpg] of a spoonful of water. After transferring it, the first glass will contain [image file=image_rsrc3TB.jpg] spoonfuls water, [image file=image_rsrc3TC.jpg] spoonfuls wine. The second glass will contain [image file=image_rsrc3TD.jpg] spoonfuls of water, [image file=image_rsrc3TE.jpg] spoonfuls of wine. The amount of wine in the first glass is therefore exactly the same as the amount of water in the second glass.

This being exactly the same might be an accident, but if you vary the conditions of the problem, you will find you always get equal amounts. If we begin with x teaspoonfuls of water and x teaspoonfuls of wine, the amount of wine that gets into the water still equals the amount of water that gets into the wine. Even if we make the glasses begin with unequal amounts, if we put x teaspoonfuls of water in one and y teaspoonfuls of wine in the other; and then follow the rest of the instructions as in the original problem, our calculation still shows that at the end the water in the first glass is equal in amount to the wine in the second glass.

Now this is a clear example of bad mathematical style. In a good proof, an illuminating proof, the result does not appear as a surprise in the last line; you can see it coming all the way.

This particular puzzle uses a kind of camouflage. It tells you something which you do not need to know, and by so doing distracts your attention from the real point. The unnecessary statement is ‘the mixture is thoroughly stirred up’. The essential point is that we transfer one teaspoonful of liquid from the first glass to the second, and then we bring back one teaspoonful of liquid from the second to the first. It does not in the least matter what kind of liquid. All that matters is that, at the end, each glass contains the same amount of liquid that it did at the beginning. If this is so, the first glass must have received just enough wine to compensate for the water that it has lost; and of course the water it has lost will be found in the second glass. The amounts cannot fail to be equal; no fractions, no algebra need be employed.

The general statement of the puzzle would therefore be: we have a glass of water and a glass of wine, we carry out any series of operations with these liquids such that the total amount of liquid in each glass is at the end what it was at the beginning: then the amount of water that has got into the wine must equal the amount of wine that has got into the water.

But this is so obvious that it is hardly worth saying. It is much simpler than the calculations we had to do earlier. But the range of puzzles to which it applies is far greater; you could switch teaspoonfuls of liquid backwards and forwards between the two glasses as many times as you liked, and the principle would still apply.

The investigation of a problem therefore consists of scraping away all unwanted information, until only the essential facts remain. The less you are told, the easier it is to find a solution. A general theorem rarely says anything complicated; what it does is to draw your attention to the important facts.

In elementary mathematics we have a hotch-potch of details. In higher mathematics, we attempt to isolate the various elements involved, and to study each by itself. Higher mathematics can be much simpler than elementary mathematics.

Perhaps the most famous example of simplification by generalization is Hilbert's Finite Basis Theorem. In 1868 Gordan had proved, by laborious calculations, a certain theorem which I will not here attempt to state. It aimed at showing that certain collections of polynomials, which had arisen in connexion with a particular theory, had a certain property. In 1890 Hilbert proved this result, very simply and without calculations. The advance was due to his throwing away 90 per cent of the information used by Gordan. He proved the result to hold, not merely for those particular collections of polynomials, but for any collection of polynomials whatever!

We have gone from the ridiculous to the sublime. Nothing could be more trivial and localized than the wine and water problem, nothing more profound and far-reaching than Hilbert's theorem. The fact that both can be covered by the same maxim, 'Greater generality and greater simplicity go hand in hand', is perhaps an example of the power of mathematical statements to bring widely separated objects under the same roof.

One cannot judge the importance of any mathematical investigation by the particular objects it discusses. Topology is an example. Topology is sometimes referred to as ‘the rubber geometry’ - the geometry of figures drawn on an elastic sheet. And so in a way it is; it does treat of the properties of such figures. But its importance derives from the fact that on a rubber sheet there are no fixed lengths. There cannot be any result like Pythagoras’ Theorem. We can only make remarks such as: This curve is continuous: that one is broken into two separate parts. Continuity is the basic property in topology, and topology has something to say about anything which is capable of varying gradually. As there are very few things incapable of such variation, topology has a very wide influence; it is increasingly becoming the concern both of pure mathematicians and of technicians; some most remarkable results can be proved by it. It has the utmost generality and the utmost simplicity.

UNIFICATION

All the tendencies we have so far discussed operate to enlarge the subject matter of mathematics. To explore, to discover patterns, to explain the significance of each pattern, to invent new patterns resembling those already known - each one of these activities increases the bulk of mathematics. From the practical viewpoint, it becomes extremely difficult to keep track of all the results that have been discovered; and a vast litter of unconnected theorems hardly constitutes a beautiful subject. Both as a business man and as an artist, the mathematician feels the urge to draw all these separate results together into one.

The history of mathematics therefore consists of alternate expansions and contractions. A problem occupies the attention of mathematicians; hundreds of papers are written, each clarifying one facet of the truth; the subject is growing. Then, helped perhaps by the information so painfully gathered together, some exceptional genius will say, ‘All that we know can be seen as almost obvious if you look at it from this viewpoint, and bear this principle in mind’. It then ceases to be necessary to read the hundreds of separate contributions, except for the mathematical historian. The variegated results are welded together into a simple doctrine, the significant facts are separated from the chaff, the straight road to the desired conclusions is open to all. The bulk of what needs to be learnt has contracted. But this is not the end. The new methods having become common property, new problems are found which they are insufficient to solve, new gropings after solutions are made, new papers are published; expansion begins again.

If it were possible to weld together the whole of knowledge into two general laws, a mathematician would not be satisfied. He would not be happy until he had shown that these two laws were rooted in a single principle. Nor would he be happy then; indeed he would be miserable, for there would be nothing more for him to do. But there is not the least likelihood of this state of stagnation arising. It is a property of life, a property without which life would be unendurable, that the solution of one problem always creates another. There always is, there always will be something to learn, something to conquer.

The way in which this happens can be seen from a remarkable unification which took place round about 1800, when it was found that the great majority of functions previously studied were particular cases of one very general function, the hypergeometric function. The theory of the hypergeometric function was then, and still is, a powerful device for drawing scattered pieces of information together. New ways of viewing the hypergeometric function were discovered. It was shown, among other things, that the special properties of the hypergeometric function were connected with the fact that it had three singularities - exactly what singularities are need not be explained just now. The functions with simple properties, then, were those with three singularities or less; it was with such functions that previous mathematics had been concerned. But this immediately raised the question - unsolved, to a large extent, to this day - what sort of properties would a function with four singularities have?

And so it will always be. If it could be shown that all existing mathematics was concerned with things having properties A, B, and C, mathematicians would immediately ask, what happens if an object only has some, or none of these properties? And they would be off again.

CHAPTER THREE

Pattern in Elementary Mathematics

Our stability is but balance, and wisdom lies

In masterful administration of the unforeseen.

R. Bridges, The Testament of Beauty

Let us consider pattern for a moment from the lowest possible point of view, that is, for a person who simply wants to pass an examination. The most important question for the examinee is the character of the examiner; what are the examiner's interests? what qualities is he trying to test in the examinee?

Some examiners seem mainly interested in the ability of the student to carry out routine operations. Does the student know his multiplication tables? Can he use logarithms? Can he use any one of a hundred other stock methods? Doubtless it is necessary to test the student's knowledge of routine processes; but ever since being a boy I have classified such tests as dull examinations. They are very popular with the poorer teachers, since the teacher knows he has merely to drill his class on exercises 1 to 50.

Other examiners wish to encourage the more enterprising teachers, the teachers who are trying to convey not merely facts but the feel of the subject; they wish to test the enterprise, the imagination, the initiative of the students. And so they seek out problems which call for these qualities.

Some teachers feel this to be unfair; they think that pupils cannot be prepared for an unpredictable examination. But this is not true. Is a battle predictable? Is it therefore impossible to train military leaders? The training of officers is based (or should be) partly on those general principles which are common to all battles, and partly in developing initiative; the officer is placed in a variety of unforeseen situations in which he has to improvise. Exactly the same type of approach is possible in the peaceful training of mathematicians. An examination and a battle have much in common.

An examination paper may contain both routine and problem questions. How is the student to know which is which? They need not look very different.

Take for example the simultaneous equations below.

[image file=image_rsrc3TF.jpg] There is a routine method for solving such equations. If we multiply the first equation by 73, the second equation by 341, and subtract the first result from the second, we find that x must be 17,849 /65,067. This answer does not simplify; there can therefore be no way of avoiding arithmetical calculations. An examiner who set such a question could merely be interested in seeing whether the students knew the routine method, and had the persistence and.the accuracy to carry through the arithmetic. (I assume the exact answer, as a fraction, is wanted. If only an approximate answer is required, the labour can, of course, be reduced - for example, by using a slide rule.) The value of y is equally complicated.

For contrast, consider the simultaneous equations

[image file=image_rsrc3TG.jpg] The numbers here are larger, but the problem is a much easier one. The problem is not intended to test routine work. There is a very simple way into it if the examinee can find how.

What clues are there to suggest that this question calls for an imaginative attack? There are certain clues in the choice of the numbers, but the most striking and significant clue is the pattern

[image file=image_rsrc3TH.jpg] which appears on the left-hand sides of the equations.

These left-hand sides have in fact the algebraic form

[image file=image_rsrc3TJ.jpg] In looking at these expressions I find myself reasoning in the following somewhat idiotic fashion. If we were to re-christen our unknowns x and y, so that x was called y and y was called x, the expression (I) would become (II), and the expression (II) would become (I). So the expressions (I) and (II) are just as good as each other. It would therefore be unfair to do anything to (I) that we do not do to (II). Whatever steps we take ought to treat (I) and (II) alike.

What steps are there of this kind? The obvious step is to add (I) and (II). This is a perfectly symmetrical operation.

Are there any other symmetrical operations that we could do? We could of course multiply (I) and (II) together, but that would not be helpful towards solving the equations. We want an operation of the form $m(\mathrm{I}) + n(\mathrm{II})$. At first sight, it seems that to be fair we must take $m = n$. Otherwise, whichever expression gets the bigger number, the other expression can complain of injustice. There is however a possible solution. The equation $p = q$ is certainly symmetrical as between p and q. But if we try to write this equation with all the non-zero terms on one side we obtain $p - q = 0$. This appears unsymmetrical; we have taken an arbitrary choice, in deciding to assign the + sign to p and the - sign to q. But if we had taken the other decision, we should have written $q - p = 0$, which again expresses $p = q$. Accordingly, although the expression $p - q$ is unsymmetrical, the equation $p - q = 0$, being a form of $p = q$, must be regarded as symmetrical. We need not feel we are sinning against symmetry if, given two expressions of equal status, we subtract one of them from the other.

Returning to our original problem, we now have two operations that seem to have the symmetry appropriate to the problem; to add and to subtract the equations given. Carrying out these operations, we find

[image file=image_rsrc3TK.jpg] that is, $x + y = 5$ and $x - y = 1$, from which $x = 3$, $y = 2$.

Children can find real pleasure in the experience of solving such a problem. Not everyone will follow the somewhat fanciful path I have just described; but anyone who solves this problem, other than by sheer calculation, will in some way be responding to the pattern of the equations. The conquest of the problem depends on sensitiveness to pattern; this combination of a military and an artistic aspect is perhaps not realized to such an extent in any activity outside mathematics.

In adult problems, nature is the examiner. And here again it is of the utmost importance to determine whether any particular problem involves a routine slog, or whether it has some special feature that will make possible a simpler solution. Many practical problems, of course, can only be solved by routine methods; they lack pattern. This is particularly so if there is an element of randomness in them. In surveying, for example, a host of geological and historical causes have interacted to determine the positions of towns and the mountains between them. One does not expect any elegant relations between the distances on the map; calculation is inevitable. On the other hand, in a fundamental scientific problem, the structure of the atom or of the universe, one does (rightly or wrongly) expect to find an underlying simplicity; the basic theories of physics usually possess mathematical elegance; their applications to complex situations, needless to say, may not.

RECONSTRUCTING AN EXAMINER

Occasionally palaeontologists dig up a small fossilized bone and proceed to reconstruct the shape of an extinct animal. A similar activity is possible in regard to examiners, the questions set taking the place of the fossil bone.

A good examination question is not just a shapeless affair; it should contain some interesting design or some surprising result. Such questions are by no means easy to make up. Accordingly, an examiner who is doing research work will usually be on the look-out for some result that he can use in an examination question. Often, in fairly advanced work, some small piece of algebraic manipulation occurs, which can be detached from its context and set as a problem.

For example, some years ago students brought me the following question, which had been set in an examination paper, and which they found hard to solve, or at any rate to solve in a satisfying manner.

‘Prove that, if8

[image file=image_rsrc3TM.jpg] then the fractions just given are both equal to

[image file=image_rsrc3TN.jpg] This question has a very definite form, and obviously to hammer it out by a lengthy and shapeless calculation, while verifying the result, would bring one no nearer to the heart of the question. What interested me most was the question, how did the examiner come to think of this question?

The pattern of the question includes the following aspects. ac - b2= 0 is the condition for the three quantities a, b, c to be in geometrical progression. The numerator of the first fraction contains ac - b2. A similar expression occurs in the numerator of the second fraction. Down below we have expressions a - 2b + c and b - 2c + d which are associated with arithmetical progressions, a - 2b + c = 0 being the condition for a, b, c to be in A.P. Again there is a kind of rule by which the denominators could be derived from the numerators; in the third fraction, for example, we have a and d multiplied on top, added below, *i.e.* the numerator contains ad, the denominator a + d. The negative terms are similarly related; on top we have -bc, down below - (b + c). This rule applies equally well to the first two fractions; in the first fraction, for instance, -b2 is - bb, and we find - (b + b), that is, - 2b, below it.

To invent a problem so knit together is almost impossible. One does not invent such things; one stumbles upon them. I was certain that the examiner had been finding the condition for something, and these fractions had arisen in the course of the work.

The way to begin the problem was fairly obvious, to bring in a new symbol, k, for the value of the fractions. The problem then can be stated as follows.

If

[image file=image_rsrc3TP.jpg] and

[image file=image_rsrc3TR.jpg] prove

[image file=image_rsrc3TS.jpg] To bring in such a symbol k is routine procedure, when dealing with the equality of several fractions. (See, for example, Hall and Knight, Higher Algebra, Chapter 1.)

What to do next was not at all obvious to me. I tried various methods which, though they led to proofs, did not satisfy me. I continued to think about this question in odd moments, and about a week later, hit on the following approach. Equation (I) can be put in the form $ac - k(a + c) = b^2 - 2bk$. Both sides of this equation are now crying out for an extra term k^2 to complete their pattern. This will ‘complete the square’ on the right-hand side, giving $(b - k)^2$, and give $(a - k)(c-k)$. on the left. So $(a - k)(c - k) = (b - k)^2$. That is to say, equation (I) expresses the fact that $a - k$, $b - k$, $c - k$ are in G.P.

Now we have the whole thing. Equation (II) shows that $b - k$, $c - k$, $d - k$ are in G.P. So $a - k$, $b - k$, $c - k$, $d - k$ are in geometrical progression. But if we multiply together the first and fourth terms of a G.P. the result equals the product of the second and third terms. (Let the G.P. be A, AR, AR^2, AR^3. Then $A \times AR^3 = AR \times AR^3$.) So we have

$$(a - k)(d - k) = (b - k)(c - k).$$

If we multiply this out, cancel k^2 and solve the resulting linear equation for k, equation (III) results.

Our conclusion therefore is that the examiner’s researches had led him on some occasion to pose the question, ‘What is the condition that four numbers a, b, c, d must satisfy, if, by subtracting the same number from each of them, a geometrical progression can be obtained?’

The moral of this is not confined to examination questions. It is meant to support the thesis, where there is pattern there is significance. If in mathematical work of any kind we find a certain striking pattern recurs, it is always suggested that we should investigate why it occurs. It is bound to have some meaning, which we can grasp as an idea rather than as a collection of symbols. It is extremely unsatisfactory to discover a theorem, and only be able to prove it by shapeless calculations. It means that we do not understand what we have discovered.

To find the significance of an algebraic formula may take a long time; there are usually so many possible methods of attack, and no way of telling which is the true one. I find that, in dealing with such problems, my brain has a delayed action; at first I can make nothing of the question; a day, or a week, or a month later, an inspiration comes. If students are required to solve such a problem within three hours, that is, under examination conditions, a considerable element of luck is brought in. The usual way of overcoming this difficulty is to put in front of the problem a piece of bookwork, or some simpler problem that will start the mind working in the correct direction.

RECONSTRUCTING TWO AUTHORS

Even more difficult than collecting enough good questions to fill an examination paper is the task of collecting enough questions to fill a textbook. The best way to get interesting questions in algebra, I believe, would be to read a large number of research papers, written over the last couple of centuries, and take note of all the algebraic results that have been proved incidentally in the course of advanced work. The most interesting results in the elementary algebra books probably did originate in this way.

If one looks through any textbook, a certain number of examples are without pattern. They are the kind of thing anyone could easily make up; they belong to the same class of approach as the arithmetic question ‘Find 27 + 46 + 39’ or the algebra exercise Multiply 5x2 - 3x + 7 by 4x + 11’. A certain number of such routine exercises are of course necessary. Other questions are of the kind that could not just be made up on the spur of the moment, and it is always interesting to try to guess how the author arrived at his problem.

For example, in Hall and Knight’s Higher Algebra (which of course is not higher algebra in the modern sense) the question occurs, If $a = zb + yc$, $b = xc + za$, $c = ya + xb$ prove

[image file=image_rsrc3TT.jpg] This is not a difficult exercise to do, but we are not at the moment concerned with how it is solved; we are concerned with how it was composed.

Unless an enormous coincidence has occurred, it was arrived at in the following way. Mr Hall and Mr Knight were discussing, not algebra, but trigonometry. Now certain algebraic questions arise in connexion with trigonometry. In algebra, as is well known, you cannot find 3 unknowns from only 2 equations. If you have 3 unknowns and 3 equations, you can (as a rule) find the unknowns. If you have 3 unknowns and 4 equations, generally speaking, an impossibility arises. For example, you might have the equations $x = 1$, $y = 2$, $z = 3$, $x + y + z = 0$, which contradict each other. 4 equations for 3 unknowns can only be solved if the 4th equation is a mere repetition of something we could have discovered for ourselves from the other three, like, for example $x = 1$, $y = 2$, $z = 3$, $x + y + z = 6$. Then there is no trouble.

[image file=image_rsrc3TU.jpg] Figure 4

Now look at the question of solving a triangle (Figure 4), when the three sides a, b, c are given, and we are asked to find the angles A, B, C of the triangle. 3 unknowns, 3 equations will be enough; more than 3 may be an embarrassment. Trigonometry gives us eight.First of all, if we drop a perpendicular AD on to BC, we have BC = BD + DC, and this shows

(1) $a = c \cos B + b \cos C$

By dropping a perpendicular from B on to AC, or from C on to AB, we find two more equations, which I will call equations (2) and (3), but not write down here.

Three equations are quite enough to find three unknowns. From the point of view of algebra, we now have all we need. But trigonometry continues to pour its gifts upon us. The textbook calls our attention to the Cosine Formula

[image file=image_rsrc3TV.jpg] and to the corresponding expressions for cos B and cos C, equations (5) and (6).

And finally there is the Sine Rule

[image file=image_rsrc3TW.jpg] which contains two ‘equal’ signs, and gives us equations (7) and (8).

Now we know that we can find the angles of a triangle, and that these various results are not in conflict with each other. It must then be that equations (4) to (8) add no new knowledge, but are algebraic consequences of (1), (2) and (3).

This, as a matter of fact, is pretty obvious for equations (4), (5) and (6), which are simply what one gets on solving equations (1), (2) and (3) for cos A, cos B and cos C. But it is not so obvious that the Sine Rule follows. Here then is a result which is not too obvious, and will make an exercise in algebra. To get rid of the trigonometrical functions, we simply put cos A = x, cos B = y, cos C = z. As sin2 A + cos2 A = 1, we shall have to put sin2 A = 1 - x2. We square the Sine Rule equations above, and are thus able to translate the equations entirely into the language of algebra. We so arrive at the problem as stated in Hall and Knight.

In trigonometry, the cosine results are usually proved by one geometrical procedure, the Sine Rule by another. The fact that the Sine Rule is an algebraic consequence of the Cosine Formula is not stressed.

Even now, we have not given the actual algebraic reasoning for deducing one from the other. But we have seen that there must be some way of proving this result algebraically. If not, trigonometry contradicts itself.

TRANSLATION FROM ONE SUBJECT TO ANOTHER

The procedure of the previous section essentially is one of translation. We begin with a known trigonometrical fact, that the Sine Rule is not inconsistent with the Cosine Formula. We translate into the language of algebra, and find that it must be possible to deduce one set of equations from another. We have been led from a familiar fact (familiar at any rate to anyone who has done school trigonometry within the last two or three years) to an unfamiliar fact. We have not merely gained a new result; we have made somewhat sharper our view of the old result. ‘Not inconsistent with’ has been sharpened to ‘can be deduced from’.

Translation is a valuable exercise, because, before you can express a fact in a new language, you must be clear what the fact is.[9] At various places in this book we shall try to state various things in such a way that they could be explained ‘to an angel over the telephone’; to explain geometry, for instance, without drawing any diagrams or invoking any physical ideas. Geometry, in fact, can be explained purely in terms of number. You may think this strange, if you have been accustomed to think of geometry as the study of the shapes of objects. But mathematics deals with patterns; it is not concerned with the particular material in which the pattern is realized. Our ability to explain geometry to the angel means simply that we can, by means of numbers alone, exhibit the same patterns as those that occur in geometry. Like the wireless commentators, we could explain to the angel the progress of a football match, by giving the score and saying, ‘Square 7’. We could not however explain to it what being kicked on the shin felt like.

As an example of translation, I will try to translate into algebra the fact that $\cos2 \theta$ never exceeds 1.

[image file=image_rsrc3TX.jpg] Figure 5

To get away from geometrical ideas we will make use of graph paper. In Figure 5 let O be the origin, P the point (a, b) and Q the point (x, y). This gives us a triangle OPQ, and anything there is to be said about this triangle can now be said in terms of the numbers a, b, x, y, that is to say, in terms of algebra. It will be convenient to use p for the length OP, q for the length OQ, and r for the length PQ. Of course p, q, r can be expressed in terms of a, b, x, y; by applying Pythagoras to the triangles ONP, OSQ and PMQ we have the very well known resultsp2 = a2 + b2

q2 = x2 + y2

r2 = (x - a)2 + (y - b)2

If the angle between OP and OQ is called θ, we can find $\cos \theta$ from the formula 2pq $\cos \theta$ = p2 + q2 - r2. On substituting for p2, q2 and r2 from the three equations above, all the square terms cancel out, and only 2ax + 2by remains. So we have

pq $\cos \theta$ = ax + by.

p and q contain square roots, if we express them in terms of a, b, x, y; however, as we want to express the fact that $\cos2 \theta$ never exceeds 1, we shall be squaring anyway. We thus find that (ax + by).2 cannot exceed p2q2, that is, (ax + by)2 never exceeds (a2 + b2)(x2 + y2).

This is a purely algebraic result; we may express it by saying that for real numbers, (a2 + b2)(x2 + y2) - (ax + by)2 is never negative.

Why is it never negative? The above statement corresponds roughly to the trigonometrical statement that 1 - $\cos2 \theta$ is never negative (for real angles θ). But this expression is the same as $\sin2 \theta$; we are familiar with the fact that squares of real numbers are never negative.

We are thus led to expect that the algebraic expression above may very well be a square. And so it is. It is no great labour to multiply it out completely. Of the seven terms so obtained, four cancel. Those remaining are equal to (bx - ay)2. This can never be negative, and we have thus proved our algebraic result directly without any appeal to trigonometry.

Our result may be written

(a2 + b2)(x2 + y2) = (ax + by)2 + (bx-ay)2.

This identity holds for all numbers a, b, x, y.

In Chapter 4 we shall return to this identity, and see in what ways it can be generalized.

KNIGHT’S TOUR IN CHESS

While we are still thinking about translation, another example may be given, which illustrates one very important use of translation, namely, to put a problem into a form where the answer can be seen at a glance. Such translation does not change the essential pattern of a problem; from a very abstract mathematical viewpoint one might say that it did nothing at all: but for us as human beings it is most valuable, since it changes the unfamiliar into the familiar.

Drawing the graph of an algebraic function illustrates my meaning; a graph can be taken in by the eye at a glance. Here we are translating from algebra to geometry. Very often we translate in the opposite direction, from geometry to algebra; several examples will be found in this book. And we may also translate from geometry to geometry, from an unfamiliar type of problem to a familiar one.

[image file=image_rsrc3TY.jpg] Figure 6

Consider the puzzle of the knight’s tour. A knight is placed in the centre of a square board of 25 squares (Figure 6). The knight is to move in such a way that it visits each square once and once only.I find it very difficulty in thinking about this puzzle to see clearly in my mind whether or not the knight is getting into difficulties when he has made, say, a dozen moves: I cannot tell by looking at the squares still unvisited whether they will join together in a single chain of knight's moves. Can we somehow restate the problem so that we shall be able to see more easily what we are doing?

From the point of view of the knight, the squares 2 and 13 are neighbours. He can jump from 13 to 2 in a single move. But, from his point of view, 12 and 13 are not neighbours; he cannot pass from one to the other in a single move. So, if we are only concerned with the problem of the knight's tour, we can forget the actual shape of the chess board. If we want to bring out the things which are important for the task in hand, we must draw a diagram of the twenty-five squares in such a way that 2 is close to 13, while 12 is not so close. (In fact, the knight needs three moves to pass from 13 to 12.)

The resulting diagram is shown in Figure 7. If a knight can pass from one square to another in a single move, the corresponding numbers are joined by a straight line. The places where these lines cross have no significance. In a diagram drawn on paper it seems impossible to avoid such accidental crossings. They could be avoided in a three-dimensional model, with wires joining the points; these wires could be bent so as to avoid each other.

[image file=image_rsrc3TZ.jpg] Figure 7The drawing of such a diagram is not quite as simple as it looks. Some care is needed if one is to avoid a tangled network of lines. The main principle used was respect for symmetry. 13 had to be in the centre, and the others arranged round it so as to preserve the symmetry of the chess board.10

It is now very easy to choose a path for the knight. He can, for example, go from 13 to 10, then go right round the 'Inner Circle' (19, 22, 11, 2, 9, 20, 23, 16, 7, 4, 15, 24, 17, 6, 3), pass to the 'Outer Circle' and go round this (12, 21, 18, 25, 14, 5, 8, 1).

We need not go into the theory of knight's tours. The example is chosen not for the importance of the subject, but rather to illustrate how a suitable way of visualizing the essential features of a problem can simplify the task of finding a solution.

A UNIQUE PATTERN

In Chapter 10, on projective geometry, we shall meet the function (a - b)(c - d)/(b - c)(d - a). This is called the cross-ratio of the four numbers a, b, c, d. As it is rather a lengthy expression, let us refer to it as f(a, b, c, d), which is somewhat shorter.

The order of the letters a, b, c, d is important. For instance, if we wrote f(a, d, c, b) that would mean

(a - d)(c - b)/(d - c)(b - a),

which is not by any means the same as f(a, b, c, d). In fact, as you can easily see, it is the reciprocal of f(a, b, c, d). If f(a, b, c, d) = x, then f(a, d, c, b) = 1 /x.

Altogether, there are 24 orders in which four letters can be written, so that there are 24 different expressions we can form, by taking a, b, c, d in different orders; for example f(a, c, d, b), f(d, c, b, a), f(d, b, a, c). If you have the patience to write all 24 expressions out, you will find that they are not all different. 6 different values occur, each of them 4 times. You can check this algebraically; or, if you like, you can take particular values, say a = 0, b = 1, c = 2, d = 3. By taking the numbers 0, 1, 2, 3 in different orders, you can get the values - [image file=image_rsrc3U0.jpg] - 3, [image file=image_rsrc3U1.jpg] 4, [image file=image_rsrc3U2.jpg] for the function; each value can be got in four different ways, for example

[image file=image_rsrc3U3.jpg] The six values so found are all connected. If we take any one of them, and turn it upside down (take the reciprocal) we get another value in the set; if we take any value and subtract it from 1, we get another value in the set. For instance, if we start witch - [image file=image_rsrc3U4.jpg] and turn it upside down we get - 3. Subtracting - 3 from 1 gives 4. Turn that upside down and get [image file=image_rsrc3U5.jpg] Subtract that from 1 to get [image file=image_rsrc3U6.jpg] and turn that upside down to get [image file=image_rsrc3U7.jpg]

Thus, if we are given any one value, we can find the others. If we start with x and turn it upside down we get 1 /x. Subtracting from one gives 1 - 1 *x, which may be written (x - 1)* x. This turned upside down gives x/(x - 1). Subtract from 1 to get 1/(1 - x) and finally turn this upside down to end with 1 - x.

You might think that by turning upside down and subtracting from 1 a few more times you might be able to get some new expressions. But you cannot. However many times you perform these operations you will still stay within the family circle of x, 1/x, 1 - x, 1/(1 - x), (x - 1)/x and x/(x - 1).

If then you were told that f(a, b, c, d) had the value 5 say, you would know that f(a, d, c, b) was [image file=image_rsrc3U8.jpg] and f(a, c, b, d) was -4 (because 1 - 5 = -4) without needing to be told what a, b, c, d were. For naturally the single equation f(a, b, c, d) = 5 is far from fixing the four quantities a, b, c, d.

The interest of this fact lies in its uniqueness. No other function does what this one does. Here, of course, we rule out trivial amendments to the function itself. For instance, the function 3f(a, b, c, d) - 2 has the same kind of property. But we regard this as the old function in disguise rather than as a new function.

Again, there are certain comparatively trivial ways of getting a similar property. We might consider a symmetric function, say ϕ(a, b, c) = a + b + c. The sum of three numbers does not depend on the order in which they are written. So, if we are told ϕ(a, b, c) = 5 we know that ϕ(a, c, b) = 5 too, and all the other ways of writing a, b, c would give 5 too. But this is too obvious to be interesting.

Slightly less obvious are functions such as

F(a, b, c) = (a - b)(a - c)(b - c).

If we exchange any two letters, this function changes sign. Thus

F(a, b, c) = - F(a, c, b) = - F(c, b, a) = - F(b, a, c) = F(b, c, a) = F(c, a, b).

It has only the two values x and - x.

However many letters are involved, we can always construct symmetric functions, like ϕ, which remain unchanged however the letters are shuffled, and functions like F, which only take two values when the letters are shuffled.

But if we want to have more than two values in the set, and to be able to predict all the values as soon as we are told one of them, we must work with four letters (such as a, b, c, d above), and our function must be the cross-ratio f(a, b, c, d) or that function thinly disguised. In this sense, the number 4 and the cross-ratio f(a, b, c, d) are singled out from all other numbers and functions.

An interesting thing is that the uniqueness of the function f(a, b, c, d) can be proved without making any calculations whatever, that is to say, by a conversational type of argument. The proof is quite short, and depends on the theory of groups.

A unique function is nearly always worth studying. If certain towns can only be reached by passing a particular bridge, traffic on that bridge is likely to be heavy. In the same way, if certain properties are possessed only by one particular function, any problem which involves those properties must be solved by means of that function. The cross-ratio f(a, b, c, d) does in fact occur in many branches of mathematics.

You may find it interesting - if you have not already done so - to verify, by elementary algebra, the property stated earlier in this chapter that if f(a, b, c, d) = x then the 24 cross-ratios formed with a, b, c, d are the functions of x stated earlier. This involves a fair amount of work. Later, in Chapter 10, when we have learnt the geometrical significance of the cross-ratio, we shall be able to prove these results with very much less labour. You will appreciate the help that geometry gives to algebra very much more if you have worked at any rate a number of these verifications out for yourself by the more elementary method.

Incidentally, for anyone learning algebra, it is often more instructive to work one problem by three or four different methods than to work out three or four different problems. Working the same problem by different methods, one has a chance of learning how these methods compare with each other in shortness and efficiency. Thus an experienced outlook is built up.

CHAPTER FOUR

Generalization in Elementary Mathematics

As mathematics passed the year 1800 and entered the recent period, there was a steady trend towards increasing abstractness and generality…. Abstractness and generality, directed to the creation of universal methods and inclusive theories, became the order of the day.

E. T. Bell, Development of Mathematics